Analytical Chemistry Cumulative Exam
Friday, September 11, 2015

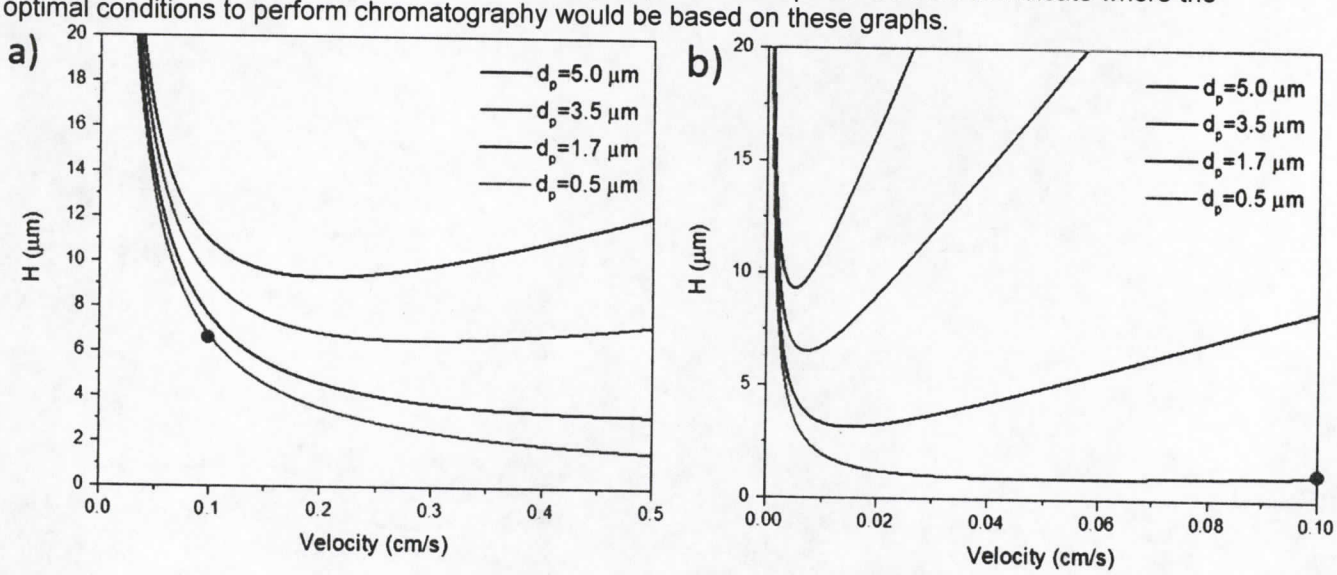
RE: Submicrometer particles and slip flow in liquid chromatography.

The journal, *Analytical Chemistry*, has regular Feature articles. For those wishing to be Ph.D.-level analytical chemists, it is critically important that you stay on top of the literature published in this journal. In particular, the Features articles describe up and coming research in the field of analytical chemistry. For your professional development, it is important to be reading this journal on a regular basis.

1. (20 points) The article begins discussing why macromolecular separations are more difficult than small molecule separations. Explain and give examples regarding why macromolecular separations are far more challenging than small molecule separations.

2. (20 points) The article describes the van Deemter equation. While Eq 1 in the article gave a more specific form of the van Deemter equation, the general form of the van Deemter equation is: $H = A + B/v + Cv$. Define each of these terms and explain how they affect a separation or what their physical and/or chemical significance is during a separation.

3. (30 points) Figure 2 from the paper is shown below. The color in these plots does not show, but from top to bottom in each graph the curves are for particles that are 5, 3.5, 1.7 and 0.5 μm in diameter. Provide an interpretation of this data for the small molecules (Graph A) vs. the macromolecules (Graph B) with a diffusion coefficient that is 10 times slower (less) than in Graph A. Be sure to indicate where the optimal conditions to perform chromatography would be based on these graphs.



4. (20 points) A significant part of the article discusses the role of extracolumn effects and their contribution to band broadening. Give examples of extracolumn effects. Using appropriate chemical or physical terms explain how extracolumn effects influence a high-theoretical plate separation.

5. (10 points) The remaining discussion in the article is about the phenomenon of the "no-slip" boundary conditions. Velocity is only zero if the mobile phase and stationary phase interactions had the same strength as mobile phase molecules to themselves. What would be the typical mobile phases used for a protein separation using a C4 or C18 (reversed-phase) material? What are the intermolecular forces that would be expected among these molecules? Comment on their thermodynamics or other important features such as bond strengths.